Seq66 on Xubuntu 26.04 – Evolution UC-33 Port-Mapping Fix

Technical report for Mr. Ahlstrom, Seq66 developer
Project directory: /media/hartmut/Daten/Synthesizer/seq66-master
Build command: ./work.sh --make
Executable tested: ./build/cc/Seq66qt5/qseq66

1. Problem

Seq66 reported an unavailable MIDI output for the Evolution UC-33, including the message "Unavailable output bus 8". ALSA detected the device, so the investigation focused on whether the ALSA connection failed or whether Seq66 failed to map the port name to the corresponding bus.

2. Device and ALSA observations

- USB ID: 0a4d:00f5; USB product string: "UC-33 USB MIDI Controller".
- The USB product string itself has no trailing whitespace.
- ALSA exposed the client as "UC-33 USB MIDI Controller"; the MIDI port names shown by ALSA had a trailing space.
- ALSA port 36:0 had RWeX capabilities and was suitable for input/output; port 36:1 was input-only.
- /proc/asound/card5/midi0 listed Output 0, Input 0, and Input 1, and showed non-zero MIDI traffic counters.
- A temporary diagnostic around snd_seq_connect_to() showed a return value of 0 for destination 36:0. The ALSA connection call itself succeeded.

3. Root cause

The saved [midi-clock-map] entry for the UC-33 was "UC-33 USB MIDI Controller MIDI" (no trailing space). At runtime, the corresponding nickname in m_clocks was "UC-33 USB MIDI Controller MIDI " (with a trailing space).
portslist::bus_from_nick_name() compared these strings exactly. The mismatch caused the lookup to fail and return null_buss() (255). The observed problem was therefore a port-name mapping mismatch, not a failure to establish the ALSA connection.

4. Source change

File: libseq66/src/play/portslist.cpp
In portslist::bus_from_nick_name(), the requested nickname and each stored io_nick_name are copied to local strings and passed through the existing rtrim() helper before comparison. The original port-name strings are not modified.

```
bussbyte
portslist::bus_from_nick_name (const std::string & nick) const
{
   bussbyte result = null_buss();
   std::string normalized_nick = nick;
   rtrim(normalized_nick);

   for (const auto & iopair : m_master_io)
   {
      std::string normalized_name = iopair.second.io_nick_name;
      rtrim(normalized_name);

      if (normalized_nick == normalized_name)
      {
```

```
            result = iopair.first;
            break;
        }
    }
    return result;
}
```

The existing util/strfunctions.hpp include provides rtrim(); no additional include was needed.

5. Build and cleanup

- The build completed successfully with ./work.sh --make (“Release build in build/cc succeeded.”).
- Temporary diagnostic fprintf() statements were removed from seq_rtmidi/src/midi_alsa.cpp, libseq66/src/play/clockslist.cpp, and libseq66/src/play/performer.cpp.
- A final build after removing the diagnostics also succeeded.

6. Verification

- Seq66 starts without the previous unavailable-output error.
- Both UC-33 input entries can be enabled in MIDI Input preferences, and the settings persist after restarting Seq66.
- In MIDI Clock preferences, the UC-33 could be changed from Disabled to On:Pos; this setting also persisted after restart.
- Port mapping and persistence of these settings were therefore verified.
- Actual MIDI clock transmission at the physical UC-33 MIDI output has not yet been independently verified; that would require a separate MIDI data test.

7. Summary

The observed Seq66 issue was caused by trailing-whitespace differences in MIDI port nicknames. Trimming trailing whitespace in bus_from_nick_name() fixed the mapping failure in the tested setup. The build is clean, temporary diagnostics have been removed, and the UC-33 settings persist across restarts.

This report describes a local source change and its observed results; it does not imply that the fix has been merged into the official Seq66 source tree.

Suggested message to Mr. Ahlstrom

Dear Mr. Ahlstrom,

I investigated the unavailable UC-33 output bus in Seq66 on Xubuntu 26.04. The ALSA connection to port 36:0 succeeded, but the saved MIDI-clock map nickname lacked a trailing space that was present in the runtime ALSA port nickname. Since bus_from_nick_name() used an exact string comparison, the lookup failed. Trimming trailing whitespace from both strings before comparing them resolved the observed mapping error. The build succeeds, Seq66 no longer reports the previous error, and the UC-33 input and MIDI Clock preferences persist after restart. Actual clock transmission at the physical MIDI output has not yet been separately tested. The attached report documents the change and verification.

Best regards,